
CAPSTONE PROJECT REPORT

Project Title: Intelligent E-Commerce Competitor Pricing & Analytics System

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1. Executive Summary

In today's highly competitive online retail industry, pricing plays a major role in gaining customers and staying ahead of competitors. However, manually monitoring competitor prices is time-consuming, inefficient, and often leads to inaccuracies.

This project focuses on automating the process of extracting competitor product information from Books to Scrape (a practice e-commerce website). The collected data is then cleaned, organized, and transformed into a structured format for analysis. After preprocessing, Machine Learning techniques are applied to predict and recommend optimal pricing strategies based on the extracted product features.

The final outcome of this project is an interactive Streamlit Dashboard, which enables stakeholders to filter and explore products, estimate prices using AI-based predictions, and download the cleaned dataset for further business intelligence and decision-making.

2. Problem Statement

2.1 The Challenge

In the competitive e-commerce environment, retailers often face difficulty in deciding the fair market price of books. Pricing is influenced by factors such as customer perception (star ratings) and product scarcity (stock availability). Without proper analysis, retailers may either overprice products (losing customers) or underprice them (reducing profit margins).

2.2 The Solution

To address this issue, this project proposes an end-to-end automated data pipeline that performs the following tasks:

1. Extracts real-time product information by scraping data from competitor websites.
2. Cleans, preprocesses, and converts the collected unstructured data into a structured dataset suitable for analysis.
3. Uses a Random Forest Regressor model to predict book prices based on important features such as rating and availability.
4. Presents the insights through an interactive web-based dashboard, allowing users to explore the pricing trends and market conditions effectively.

3. Methodology & Technical Approach

This project was developed by following a structured Data Science lifecycle, based on the CRISP-DM framework, ensuring that each stage of the pipeline was executed in an organized and systematic manner.

3.1 Phase 1: Data Acquisition (Web Scraping)

Data Source : books.toscrape.com

Tools Used : Python, Requests, BeautifulSoup

3.1.1 Approach:

A web scraping script was developed to automatically collect competitor book information from the website. The script was designed to iterate through the entire catalogue, covering 50 pages of listings. Pagination was handled dynamically, allowing the dataset to be collected without manual intervention.

A total of 1,000 book records were successfully scraped. The following attributes were extracted for each product:

- 1.Book Title
- 2.Price
- 3.Star Rating
- 4.Availability Status / Stock Details
- 5.Product URL

3.2 Phase 2: Data Cleaning & Preprocessing

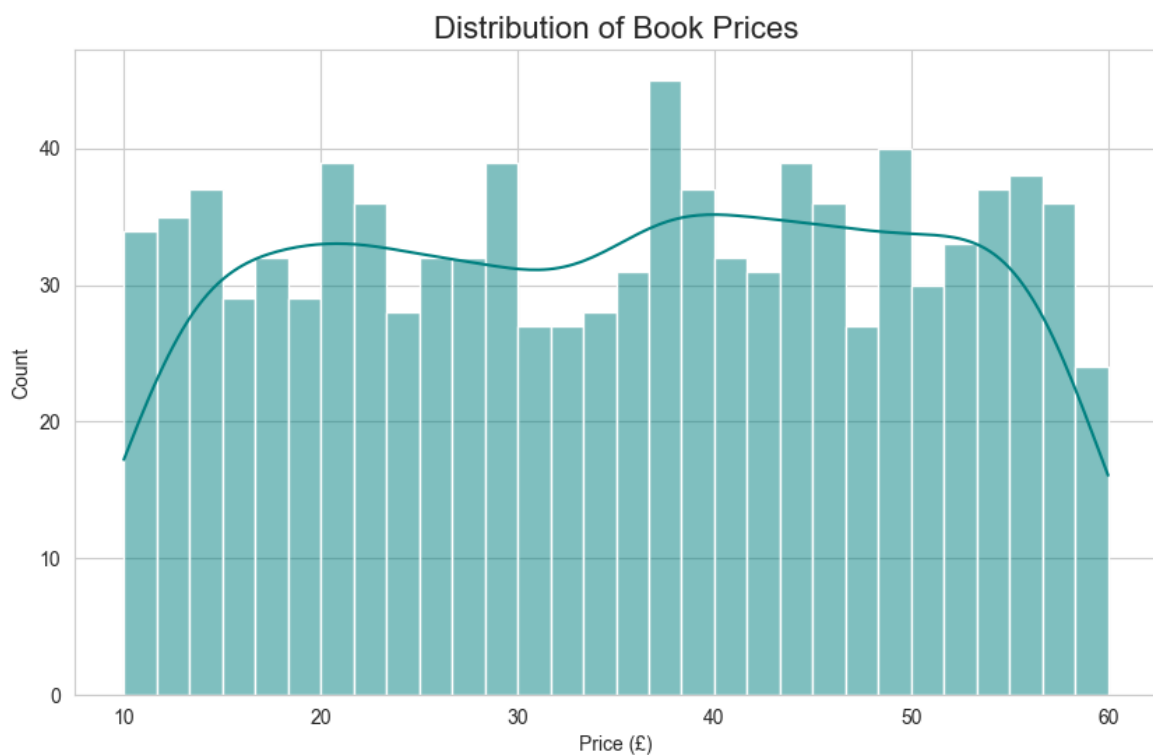
Since raw web-scraped data is usually inconsistent and unstructured, preprocessing was performed using Pandas to convert the dataset into an analysis-ready format. The major cleaning steps included:

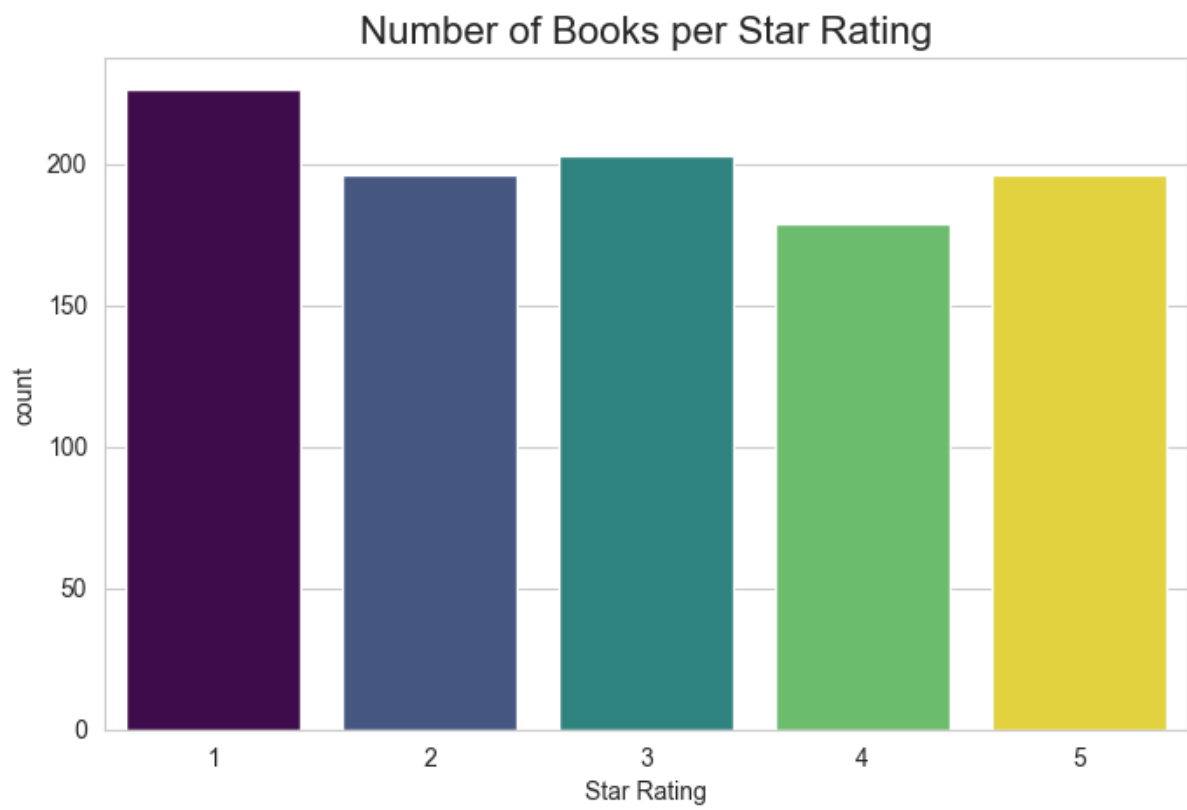
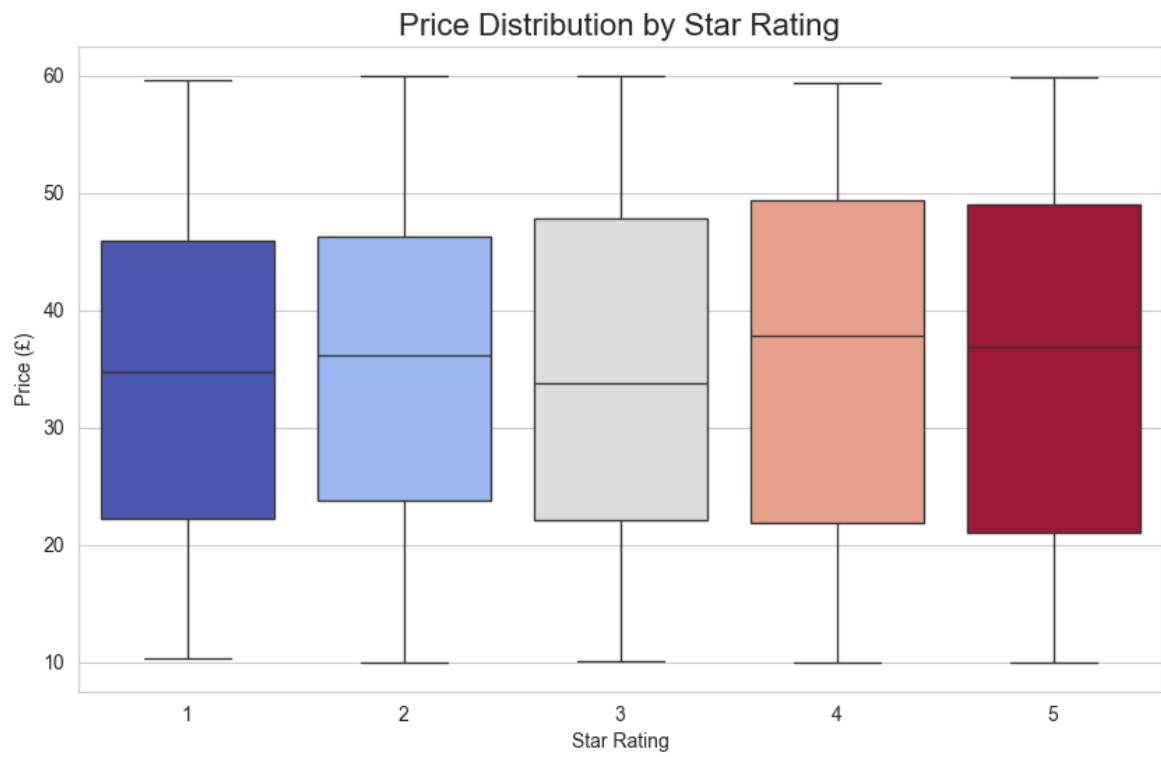
- 1.Price Normalization: Currency symbols (£) were removed and prices were converted into numeric *float* values.
- 2.Rating Conversion: Text-based ratings such as "*Three*" and "*Five*" were mapped into integer values ranging from 1 to 5.
- 3.Stock Extraction: Regular Expressions (Regex) were used to extract the numeric stock count from strings such as "*In stock (22 available)*".
- 4.Data Quality Validation: The dataset was checked for missing values and duplicates, and it was confirmed that there were 0 null values and 0 duplicate records.

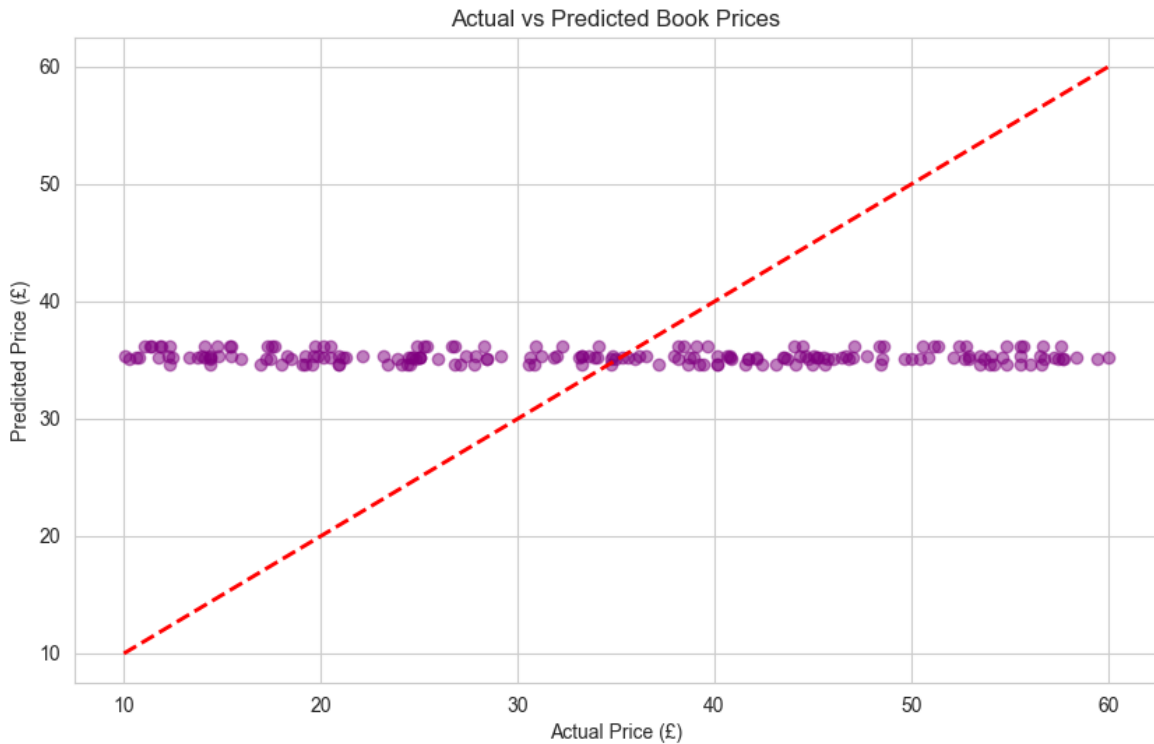
3.3 Phase 3: Exploratory Data Analysis (EDA)

Exploratory analysis was conducted to understand the overall market structure and identify meaningful patterns within the scraped dataset. Key findings include:

- 1.Price Distribution: The average price of books in the dataset was found to be approximately £35.07, with most values spread fairly evenly between £10 and £60.
- 2.Rating vs Price Relationship: The correlation between star rating and price was extremely low (0.02), indicating that competitor pricing does not strongly depend on rating. This suggests the presence of a relatively fixed pricing strategy rather than a value-based one.
- 3.Dataset Completeness: All 1,000 books were successfully extracted and accounted for, confirming that the scraping pipeline was reliable.







3.4 Phase 4: Machine Learning Model Development

To predict book prices based on available features, a Random Forest Regressor model was implemented. Random Forest was chosen because it performs well on non-linear datasets and reduces overfitting by combining multiple decision trees.

3.4.1 Model Details:

1. Algorithm Used: Random Forest Regressor (Ensemble Learning Method)
2. Input Features (X): Star Rating, Stock Availability
3. Target Variable (y): Book Price
4. Train-Test Split: 80% Training, 20% Testing

3.4.2 Model Performance:

1. MAE (Mean Absolute Error): ~£12.70
2. RMSE (Root Mean Squared Error): ~£14.59

Although the prediction accuracy is moderate, this is mainly due to the limited nature of the dataset and the randomness of pricing patterns in the scraped sample. However, the overall ML pipeline is fully functional

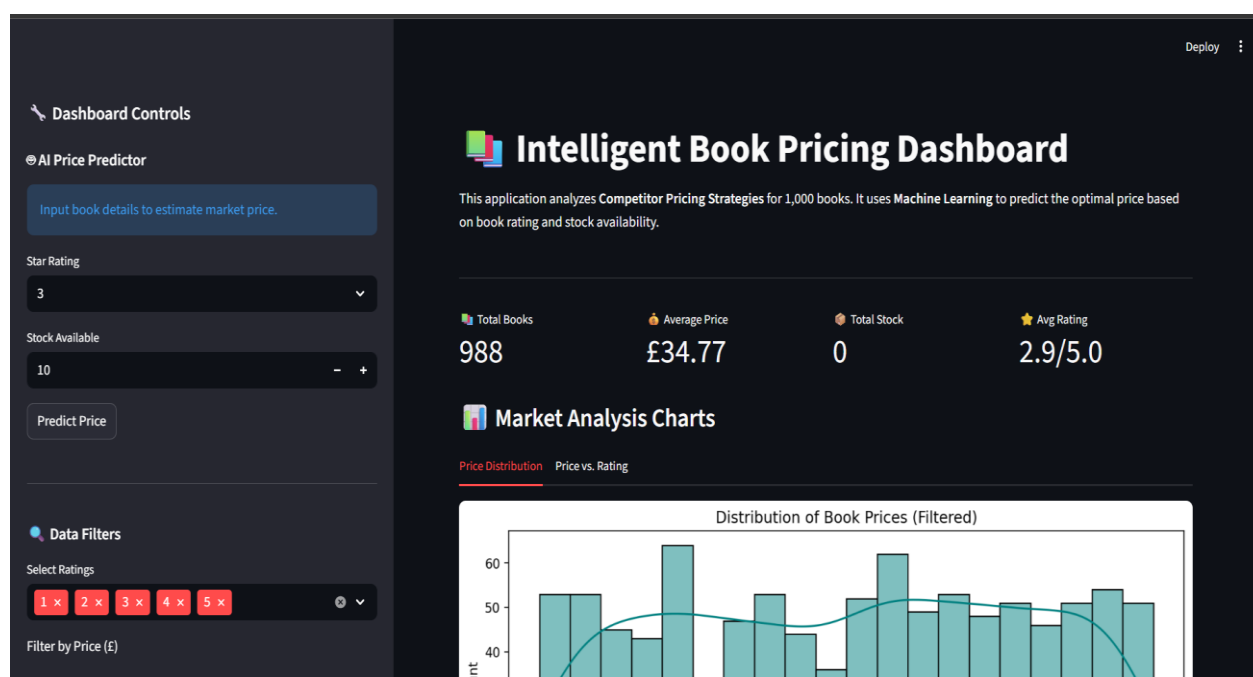
and can be easily deployed and improved further when applied to real-world competitor pricing data with stronger feature relationships.

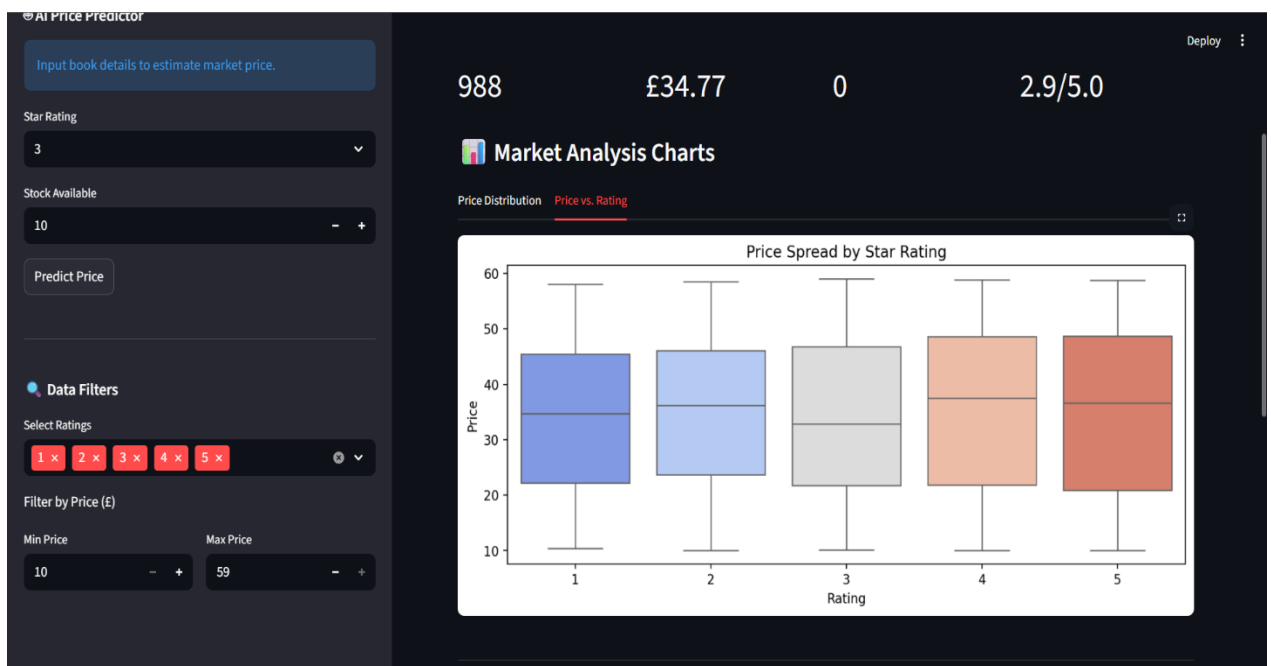
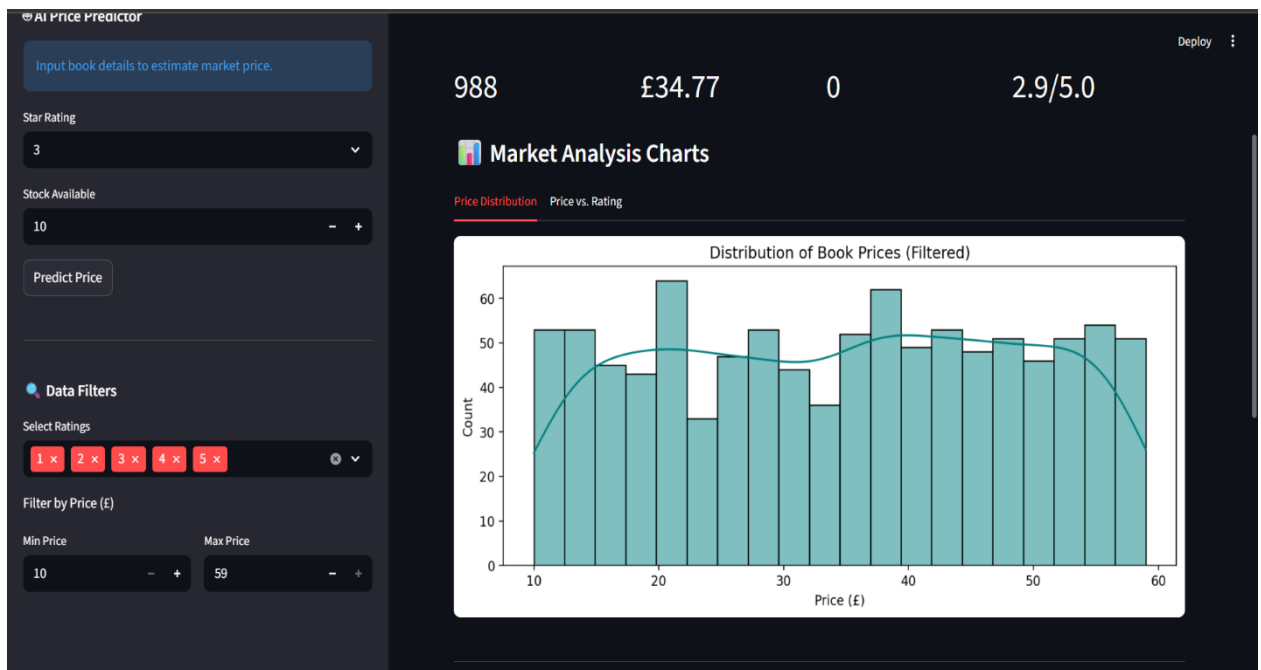
4. Final Solution: The Intelligent Dashboard

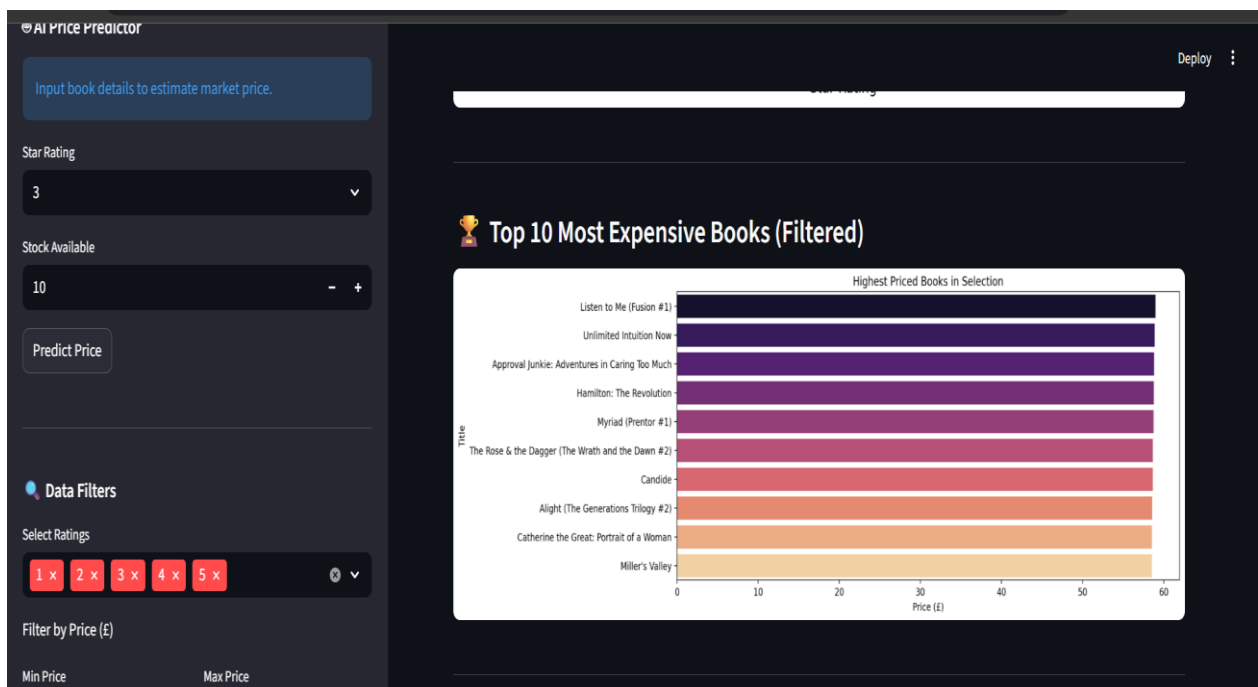
To make the results of this project easily accessible and user-friendly, an interactive web application was developed using Streamlit. This dashboard serves as the final deliverable, allowing stakeholders and analysts to explore competitor pricing data and gain insights without requiring technical knowledge of coding or data science tools.

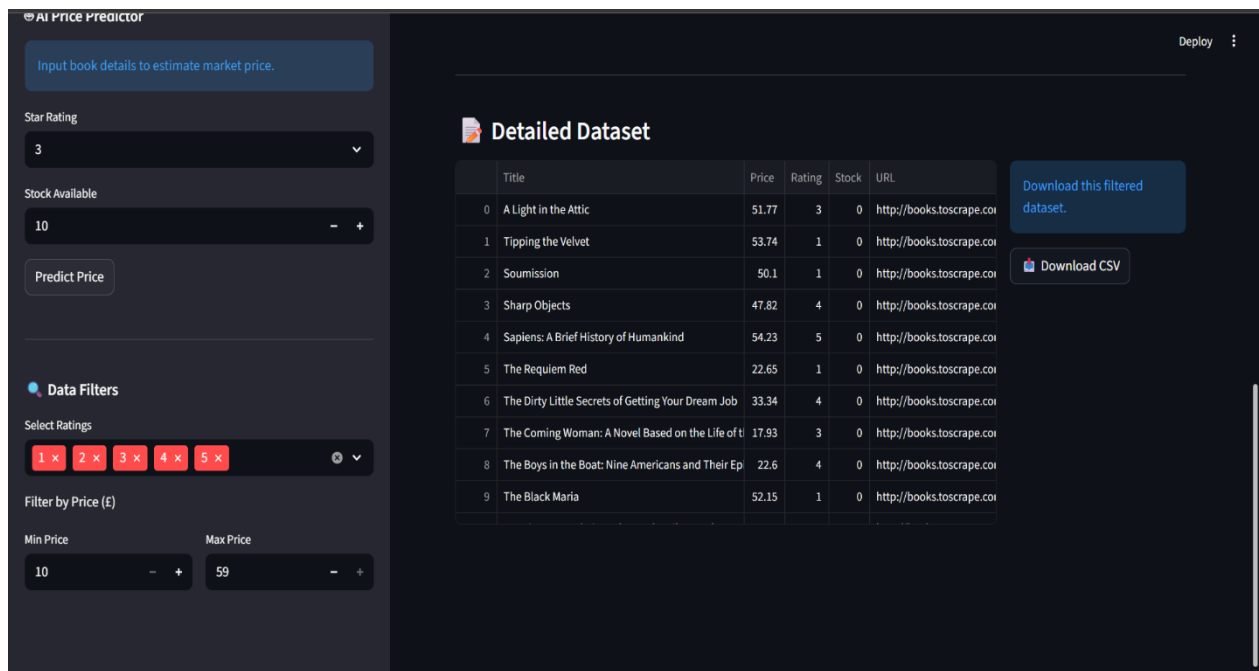
4.1 Key Features

- 1.AI Price Predictor:** The dashboard includes a sidebar-based prediction tool where users can input parameters such as Star Rating and Stock Availability. Based on these inputs, the trained machine learning model predicts the estimated market price in real time.
- 2.Dynamic Filtering:** Users can interactively filter the product catalogue by selecting a minimum and maximum price range, as well as filtering books based on star ratings. This helps in identifying specific market segments quickly.
- 3.Data Export Option:** A built-in Download CSV feature allows users to export filtered datasets. This is useful for analysts who want to perform offline reporting, create visualizations, or generate business intelligence insights using external tools.









5. Challenges & Solutions

During the development of this project, several practical challenges were encountered while handling real-time web-scraped data. Since the extracted data was not in a clean tabular format, additional preprocessing steps were required to ensure accuracy and smooth analysis. The major challenges and their solutions are described below:

5.1 Challenge 1: Unstructured Availability Information

The Availability column contained mixed text values such as *"In stock (19 available)"*, which could not be directly used for numerical analysis or machine learning.

5.1.1 Solution:

To extract only the stock count, a Regular Expression (Regex) approach was implemented. The function `re.findall(r'\d+', text)` was used to isolate the numeric value from the string and convert it into an integer format. This made the availability feature usable for both analysis and model training.

5.2 Challenge 2: Price Column Containing Currency Symbols

The Price column included the currency symbol (£), which caused issues during numerical operations and statistical calculations.

5.2.1 Solution:

The symbol was removed using the Pandas `.str.replace()` method, and the cleaned values were then converted into float format. This ensured that price-related computations and machine learning processing could be performed without errors.

6. Conclusion

This project successfully demonstrates how automation and data science can enhance decision-making in the e-commerce domain. By extracting competitor information through web scraping and transforming unstructured web data into a structured dataset, the project enables meaningful analysis and price prediction using machine learning techniques.

The final Streamlit-based dashboard provides an interactive platform that reduces the need for manual competitor research, saves significant time, and supports data-driven pricing decisions. Overall, the complete pipeline from data acquisition to model deployment proves to be efficient, practical, and scalable.

Additionally, the approach can be extended further by integrating more features such as product categories, reviews, and real time market updates. With proper permissions and compliance, this system can also be adapted to work with real-world platforms such as Amazon or Flipkart for advanced competitive pricing analysis.